and Information Technology
Examination Office

ونقـانـة المعـلـومـات
لجنة الإمتحانات

امتحانات الدور الاول ابريل 2026 م
المستوى الرابع- الدفعة (11)- هندسة البرمجيات

اسم المادة: تصميم النظم المدمجة	التاريخ :2026/5/7 م
استاذ المادة : د.كمال بشير	الزمن :3 ساعات

اجب عن جميع الاسئلة

Question 1. Definitions :Define the following terms:

1. Embedded System
2. Digital System
3. Design Metrics
4. NRE Cost
5. Unit Cost
6. Latency
7. Throughput
8. General-Purpose Processor
9. Application-Specific Processor
10. Single-Purpose Processor
11. Hardware/Software Co-Design
12. System on a Chip (SoC)
13. Design Productivity Gap
14. Reactive System
15. Real-Time System
16. Abstraction (in design)
17. Allocation
18. Partitioning
19. Communication Synthesis Shared variable access between concurrent
20. Mythical Man-Month

Question 2. True/False

Indicate whether the statement is true (T) or false (F):

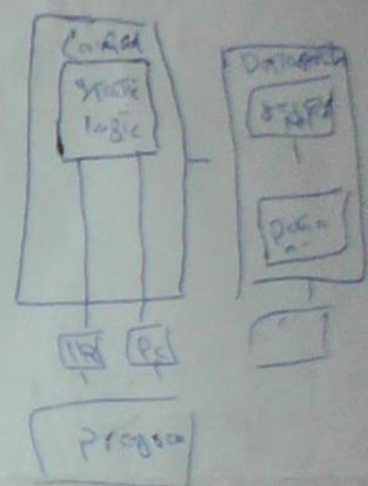
1. Embedded systems are always part of a larger system ✗
2. NRE cost is a recurring expense in manufacturing.
3. Throughput measures tasks completed per second. ✓
4. General-purpose processors are optimized for specific applications ✗
5. A digital camera is an example of a tightly constrained embedded system ✓
6. Time-to-market refers to the time required to fix a product after release ✗
7. Power consumption is a critical design metric for embedded systems.
8. SoC stands for "Software on Chip." ✗
9. Dynamic scheduling minimizes context-switching overhead. ✗
10. The design productivity gap has narrowed in recent years. ✗
11. Shared memory is a type of inter-PE communication scheme ✗
12. Functional validation is performed only at the final design stage. ✗

13. Co-design involves integrating hardware and software components. ✓
14. Unit cost decreases as production volume increases.
15. MPEG decoders are examples of embedded systems. ✗
16. Tightly-constrained systems prioritize flexibility over cost.
17. The "mythical man-month" refers to delays caused by poor team management. ✓
18. Embedded systems are rarely used in medical devices.
19. Static scheduling relies on runtime decisions. ✗
20. Design metrics often compete with each other. ✓

Question 3. Direct Answers

Answer briefly:

1. List three key technologies for embedded systems.
2. Name two components of an embedded system's I/O interface.
3. What does NRE stand for?
4. Give two examples of design metrics.
5. What is the primary advantage of application-specific processors?
6. How does the "market window" affect time-to-market?
7. What are the three main types of processors?
8. What is the difference between latency and throughput?
9. Name two tools used for analysis and validation in co-design.
10. What is the purpose of communication synthesis?



Question 4. Discussion/Comparison/Drawing

1. Compare general-purpose, application-specific and single-purpose processors.
2. Discuss how delays in time-to-market impact product revenue.
3. Draw a block diagram of an embedded system's components and label them.
4. Explain the design productivity gap and its implications.
5. [illegible]
6. Describe the co-design methodology with a flowchart.
7. Analyze the trade-offs between NRE cost and unit cost for high-volume production.
8. Discuss the role of sensors and actuators in embedded systems.
9. Explain the concept of "abstraction" in system design with an example.
10. Compare shared-memory and message-passing communication schemes.

Good luck

Analysing → Allocated - Prototyping - Scheduling → communication - implementation [HW, SW].